Response to the Referees

Manuscript: **CQG-116364**, *Low-lying quasinormal modes of Kazakov–Solodukhin black holes from Chebyshev and continued-fraction methods*

We thank the editor and referees for their careful and constructive reports. The manuscript has been extensively revised. We have derived the minimally coupled scalar equation, replaced the former lapse-substitution axial proxy with a gauge-invariant sourced odd-parity formulation under an explicitly stated inverse-Cowling closure, regenerated the complete axial catalogue, documented the spectral candidate-selection algorithm, added continued-fraction depth and Taylor-order convergence studies, replaced condition-number reporting with scale-aware backward errors, added external high-precision axial comparisons, documented all 18 Schwarzschild branch seeds, audited the positive-imaginary raw spectrum before damped-mode filtering, clarified the confidence hierarchy of the overtone catalogue, and added the requested physical and graphical diagnostics. The minimally coupled scalar sector remains the least assumption-dependent KS result, and the methodological contribution is an audited low-lying-mode workflow rather than a new high-overtone solver.

We have revised the title to reflect more accurately the manuscript's low-lying spectral scope and its use of two complementary numerical methods. The new title is *Low-lying quasinormal modes of Kazakov–Solodukhin black holes from Chebyshev and continued-fraction methods*.

Page and line references below refer to the blue, line-numbered marked manuscript.

Referee 1

Referee 1, Comment 1

> My first concern is about the theoretical setup in Sec. 2.1. The spacetime form in Eq. (1) is introduced without any reference or derivation, although it is not, at least to me, a completely standard expression. The same issue occurs for the effective potentials in Eqs. (3) and (4). The authors simply write down these potentials, but do not explain where they come from. This is particularly important for the axial-gravitational case, because the potential is later used more like a lapse-deformed Regge-Wheeler potential than a perturbation equation derived from a full gauge-invariant analysis. The authors should either provide a derivation or clearly cite the original source and state the assumptions behind these equations.

Response: We agree that the provenance and assumptions were previously underexplained. Section 2.1 now cites the original Kazakov–Solodukhin construction and a later paper using the same lapse convention, explains the areal-radius and deformation-parameter notation without introducing an unused renormalized coupling, states the coordinate domain $r \geq a$, derives $r_h = \sqrt{4M^2 + a^2}$, and displays the Schwarzschild limit. We now derive the scalar potential from $\Box\Phi = 0$ using $\Phi = e^{-i\omega t} Y_{\ell m} \Psi / r$. For the axial equation, we went beyond retaining a phenomenological disclaimer. Following Gerlach–Sengupta, Gundlach–Martínez-García, and Karlovini, the revision maps the harmonic and volume-form conventions; defines the odd gauge transformation, invariant one-form k_a, invariant curl Π, and invariant source current L_a; and displays the general sourced master equation before imposing a closure. We corrected the foundational citation to Physical Review D 19, 2268 (1979) and give the exact equation numbers used. Setting $L_a = L = 0$ both removes the source and satisfies its conservation equation. This is stated as an additional inverse-Cowling closure motivated by, but not uniquely prescribed by, the original KS mode split. The resulting potential contains the effective density and radial pressure and differs from simple lapse substitution for $a > 0$. We also compare the same potential and 18 overlapping frequencies with the public 2026 Batic–Dutykh–Sukaiti calculation at an exactly cited repository commit.

Changes in the manuscript: Section 2.1, pages 3–6, lines 102–169; axial external validation and stability analysis, pages 13–14, lines 327–367.

Referee 1, Comment 2

> The Schwarzschild benchmark in Sec. 3.1 also needs clarification. The authors quote a "known" scalar Schwarzschild fundamental frequency, but it is not clear where this number comes from or how it was obtained. Similarly, when they say that "the Leaver-style solver gives" a certain value, the truncation order, convergence criterion, and numerical stability are not specified. For the scalar Schwarzschild fundamental mode, many standard methods can reach very high precision once the order is high enough. Therefore, reproducing this number does not by itself show the advantage of the present workflow. The authors need to explain more clearly what is actually improved by their method.

Response: We now identify the external value $M\omega = 0.483643872211 - 0.096758775978i$ as the spin-zero, $\ell = 2, n = 0$ Schwarzschild result reported by Cavalcante and Carneiro da Cunha (Table I, page 6), with $M = 1$, $e^{-i\omega t}$, and $\operatorname{Im}\omega < 0$. The manuscript now also defines $\Delta_{\mathrm{rel}}(\omega_1, \omega_2) = |\omega_1 - \omega_2|/|\omega_2|$, where ω_2 is the reference value. The revised methods give the actual implementation settings: Taylor order 96, continued-fraction/Gaussian-elimination depth 240, IEEE-754 double precision, SciPy/MINPACK hybrid root finding, root tolerance 10^{-11}, at most 1000 function evaluations, the residual thresholds, branch-matched $N = 32$ initial guesses, and continuation in a/M. A new depth-truncation table covers the Schwarzschild fundamental, one KS-deformed fundamental, and a KS first overtone. A separate Taylor-order table varies orders 64, 80, 96, 112, and 128 at fixed depth 320. Figure 1 now plots error from an order-128, depth-320 continued-fraction root through $N = 128$, for both the fundamental and first overtone. It exposes the overtone's high-N deterioration and supports the more precise label "cross-validated low-lying first overtones at $N = 32$." The same settings are stored in `config/leaver_revision.json`.

We also rewrote the novelty claim. Agreement with Schwarzschild is presented as a cross-discretization check, not a claim of superior precision. The contribution is the traceable chain from equilibrated generalized-eigenvalue candidates through explicit filtering, clustering, continuation, branch scoring, singular-value refinement, spectral-size convergence, polynomial backward-error diagnostics on the raw matrices, and collocation-independent continued-fraction validation. We did not manufacture a rejected-candidate example where the available data did not support one; instead, the paper states precisely that the advantage is confidence and traceability.

Changes in the manuscript: Introduction, pages 1–3, lines 20–61; root-selection algorithm, pages 8–9, lines 219–260; continued-fraction methods, pages 9–10, lines 261–285; Schwarzschild benchmark and convergence evidence, pages 10–12, lines 287–326; residual diagnostics, pages 18–19, lines 442–448; conclusion, page 24, lines 541–570.

Referee 1, Comment 3

> I am also not fully convinced by the claimed capability of the method for higher overtones. The paper mainly focuses on low-lying modes, and even the second overtone already shows larger discrepancies and requires some caution. This suggests that the present

> method may be more reliable for low-lying modes than for high-n modes. This limitation should be stated explicitly. In particular, recent Chen-Heun-type methods have already made substantial progress on the accuracy of high overtones and have obtained complete quasinormal-mode spectra with very high precision. In comparison, the present method does not seem to solve the high-overtone problem, and the authors should avoid giving the impression that it provides a general complete-spectrum method.

Response: We agree and have narrowed the scope explicitly. Fundamentals are classified as robust fundamental modes, scalar $n = 1$ modes as cross-validated low-lying first overtones at $N = 32$, second overtones as exploratory diagnostics, and higher overtones as outside scope. These tiers appear in the abstract, introduction, results, claim-hierarchy table, conclusion, and generated catalogue CSV. We also cite Chen *et al.* (2025), explain that confluent-Heun analytic continuation targets complete and highly damped type-D spectra, and state that our workflow serves the different purpose of auditing low-lying KS branches.

Changes in the manuscript: Abstract, page 1, lines 1–7; Introduction, pages 2–3, lines 54–61; catalogue discussion, pages 13–15, lines 327–367; branch-status table and limitations, pages 22–23, lines 497–518; conclusion, page 24, lines 541–570.

Referee 1, Comment 4

> Finally, I think the axial-gravitational results should be presented more carefully. Since the axial potential is not derived from a full gauge-invariant perturbation theory for the KS spacetime, these results should not be put on exactly the same physical footing as the scalar results. In my view, the scalar sector is the more solid part of the manuscript, while the axial sector is better regarded as a phenomenological extension unless a more complete derivation is supplied.

Response: We agreed with the referee's conditional phrase "unless a more complete derivation is supplied" and supplied that derivation. The revision now uses the general sourced gauge-invariant odd-parity formalism for spherical backgrounds, includes the effective-source contribution, and replaces the old lapse-substitution potential in both numerical solvers. Gauge invariance is proved at the perturbation-variable level, but is not equated with a unique KS quantum-source perturbation theory. We separately state the dynamical closure—zero invariant axial effective-source current—so the reader can see exactly what follows mathematically and what remains an assumption about the unprovided quantum nonspherical sector. This inverse-Cowling closure is our added model assumption. The scalar result remains the least assumption-dependent headline, while the axial frequencies are conditional odd-parity modes of this explicit model.

All axial frequencies, figures, tables, shifts, stability diagnostics, and external comparisons in the revised manuscript were regenerated using the inverse-Cowling effective-source potential. No accepted numerical results from the earlier lapse-substitution ansatz remain.

Changes in the manuscript: Abstract, page 1; Section 2.1, pages 5–6, lines 128–169; axial subsection, pages 13–14, lines 327–367; limitations, pages 22–23, lines 497–518; conclusion, page 24, lines 558–566.

Referee 2

Referee 2, Comment 1

> The lapse-deformed Regge-Wheeler potential introduced in Equation (4) is a helpful phenomenological toy model for testing the numerical limits of your Chebyshev-Leaver workflow. However, as correctly noted later in the discussion, it is not derived from first-principles gauge-invariant perturbations of the true quantum-corrected field equations. In a full effective field theory framework, modifications to the spacetime geometry usually alter the linearized field equations themselves, introducing extra dynamic terms beyond a simple swap of the lapse function. To ensure physical clarity for the reader, please add a brief, explicit disclaimer directly below Equation (4) in Section 2.1 stating that this sector is intentionally phenomenological and serves primarily as a numerical benchmark. This will properly align the physical claims with the excellent numerical strengths of the paper.

Response: We addressed the underlying concern more fully than the requested disclaimer: the lapse-substitution toy model has been removed. Revised Section 2.1 begins with the general sourced gauge-invariant odd-parity equation for the effective KS background, displays the density/pressure terms missing from the toy model, and gives the explicit correction relative to the old potential. Immediately after the result we impose an explicitly additional inverse-Cowling closure, whereas a theory that dynamically quantizes nonspherical source modes could add source terms or couplings.

Changes in the manuscript: Section 2.1, pages 5–6, lines 128–169.

Referee 2, Comment 2

> In Table 2, the relative error increases for the higher overtones, reaching 1.833×10^{-5} for the $\ell = 4, n = 2$ mode. Please clarify whether these higher-overtone results are intended primarily as numerical diagnostics or should be interpreted as quantitative physical predictions with the same level of confidence as the fundamental modes.

Response: They are not assigned equal confidence. The revision labels $n = 0$ as robust quantitative, $n = 1$ as validated low lying, and $n = 2$ as exploratory/diagnostic. We also separated two different error notions. The former 1.833×10^{-5} number compared the computed Schwarzschild axial root with an unevenly rounded transcribed reference, so it was not a clean solver-error estimate. We removed that literature-relative-error column. Spectral–Leaver disagreement and the uniform high-precision external Batic–Dutykh–Sukaiti comparison are reported separately.

Changes in the manuscript: Axial continued-fraction table and caption, pages 13–14, lines 327–367; branch-status statement, page 14, lines 360–367; interpretive-status table, pages 22–23, lines 497–518.

Referee 2, Comment 3

> In Section 3.4, the author mentions that the largest endpoint fractional frequency shift observed across the full fixed-mass catalog is 7.23%, occurring specifically for the scalar $\ell = 4$ fundamental branch. Since larger multipole numbers probe different regions of the effective potential barrier compared to the dominant $\ell = 2$ mode, this maximal shift is an interesting physical result. The author should include a brief physical intuition explaining

why the higher angular number shows a heightened sensitivity to the parameter a/M under a fixed mass scale.

Response: We first audited the percentages and now distinguish real-part shift, damping-magnitude shift, quality-factor shift, and complex-plane displacement. The quoted 7.23% is the $\ell = 4$ fundamental complex-plane displacement; its real-part shift is -7.27%. We then computed r_{peak}, V_{peak}, and the tortoise-coordinate curvature for $\ell = 2, 3, 4$. At $a/M = 1$, the WKB height proxies change by -7.01%, -7.13%, and -7.18%, respectively, with a similarly modest multipole dependence in curvature. The revised explanation therefore uses cautious barrier intuition and does not claim a qualitatively distinct $\ell = 4$ mechanism. Because the axial equation itself has now been replaced by the derived matter-corrected potential, the revised full-catalogue maximum is the closure-dependent axial $\ell = 2$ fundamental at 7.37%; the 7.23% statement is retained only as the scalar-fundamental comparison the referee asked us to explain.

Changes in the manuscript: Catalogue trends and new potential-peak table, pages 15–16, lines 368–391.

Referee 2, Comment 4

> In Section 3.5, the text highlights a 4.83% decrease in the quality factor $Q = \mathrm{Re}(\omega)/(2|-\mathrm{Im}(\omega)|)$ as a key physical insight for the scalar fundamental branch. To complement this discussion, the author may consider adding a small inset or an extra panel to Figure 3 or 4 that explicitly plots Q versus a/M. This will provide a direct visual complement for the spectroscopic trends described in the text.

Response: Figure 5 now contains a dedicated right panel showing $Q = \mathrm{Re}\,\omega/[2(-\mathrm{Im}\,\omega)]$ versus a/M for the robust scalar $\ell = 2, n = 0$ branch. The definition is identical in the abstract, text, and caption.

Changes in the manuscript: Spectroscopy subsection and Figure 5, page 16, lines 392–413.

Referee 2, Comment 5

> In Section 3.6, please clarify the physical significance of the endpoint $a/M = 1$. Specifically, state whether it corresponds to a physical boundary of the Kazakov-Solodukhin black hole (e.g., extremality or horizon degeneracy) or is simply a numerical cutoff adopted for the parameter scan.

Response: The revised text derives $r_h = \sqrt{4M^2 + a^2} > a$ and explicitly states that $a/M = 1$ is neither extremal nor a degenerate-horizon limit. It is the upper endpoint of the moderate deformation interval selected for this numerical study. We also clarify the conversion to the horizon-normalized convention: $a/M = 1$ corresponds to $a/r_h = 1/\sqrt{5}$, not $a/r_h = 1$.

Changes in the manuscript: Section 2.1, page 4, lines 112–114; literature-normalization discussion, pages 17–18, lines 414–438.

Referee 2, Comment 6

> To provide a more comprehensive background, please add a few brief sentences in the introduction summarizing other standard numerical methods used for calculating quasinormal modes such as the WKB approximation, Asymptotic Iteration Method (AIM), Frobenius method or direct time-domain integration etc. The discussion may be

> supported by citing representative references such as Rev. Mod. Phys. 83 (2011) 793, Phys. Rev. D 68, 024018 (2003), Eur. Phys. J. C 84, 1245 (2024), Class. Quantum Gravity 27, 155004 (2010), Eur. Phys. J. C 85 (2025) 1223, Phys. Rev. D 30, 295 (1984).

Response: We added a focused numerical-method paragraph covering WKB, direct time-domain evolution, Frobenius/Leaver continued fractions, AIM, spectral/pseudospectral eigenvalue methods, and recent confluent-Heun high-overtone work. Representative references were checked and added without turning the introduction into a general review.

Changes in the manuscript: Introduction, pages 1–2, lines 20–61.

Referee 2, Comment 7

> In the abstract, several variables and parameters are rendered as normal prose text rather than in standard math font. Please ensure that all mathematical symbols throughout the abstract are consistently formatted in math font to maintain structural uniformity with the main text.

Response: The abstract was rewritten after the scientific revisions. Every symbol and numerical expression, including N, ℓ, n, a/M, Q, and scientific notation, is now in math mode. The revised abstract also states the low-lying confidence tiers and the axial caveat.

Changes in the manuscript: Abstract, page 1, lines 1–7.

Referee 2, Comment 8

> Please use "Figure 2" instead of "figure 2" (apply this consistently throughout the manuscript).

Response: All prose figure references now use "Figure." We also audited table and equation references, ℓ notation, and the hyphenation of "gauge-invariant," "continued-fraction," "high-overtone," and "lapse-deformed."

Changes in the manuscript: Manuscript-wide editorial correction.

Verification and reproducibility

All generated tables and figures were regenerated from the revised code. The candidate-selection subsection now records the physical frequency window, non-finite-root removal, 10^{-7} clustering, row/column equilibration, 0.20 continuation gate, competing-match score, L-BFGS-B settings, and the backward-error and continued-fraction acceptance gates. Figure legends use publication terminology, including "axial inverse-Cowling." The scalar pseudospectral endpoint diagnostic was rerun for grids $81^2, 121^2, 161^2$, half-widths $0.020, 0.025, 0.030$, and $N = 32, 48, 64$; its endpoint change is positive in every test and spans 0.1002–0.1647, so the abstract now states the robust sign rather than a single window-dependent decimal. The AI disclosure now records the exact model identifier `gpt-5.6-sol`, verified from the relevant local Codex session metadata, together with the dates, detailed uses, and author-responsibility statement.

Every documented generator and test command was also run successfully from a fresh sparse clone with no pre-existing `outputs/` directory. This audit found and corrected one bootstrap issue in which the main pipeline assumed that the parent output directory already existed. The fast test suite passes 8/8 checks, including a symbolic/numerical audit of the new axial potential, and the complete validation suite passes, with the worst spectral–Leaver catalogue difference 1.202×10^{-5} on the deliberately exploratory axial $\ell = 2, n = 2, a/M = 0.5$ row. Both the clean and marked 28-page manuscripts compile independently.